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Correction of the series of exercises analysis 1

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THE REAL NUMBERS

Exercise 1 : Let x and y be two real numbers such that $0 < x \leq y$. We define

$$m = \frac{x+y}{2} \quad \text{arithmetic mean}$$

$$g = \sqrt{xy} \quad \text{geometric mean}$$

$$\frac{1}{h} = \frac{1}{2} \left(\frac{1}{x} + \frac{1}{y} \right) \quad \text{harmonic mean}$$

Show that $x \leq h \leq g \leq m \leq y$.

Solution : Let x and y be two real numbers such that $0 < x \leq y$

1) We have $x = \frac{x+x}{2} \leq \frac{x+y}{2} = m \leq \frac{y+y}{2} = y$.

So, $x \leq m \leq y$.

2) We have than $x = \sqrt{x.x} \leq \sqrt{x.y} = g \leq \sqrt{y.y} = y$.

So, $x \leq g \leq y$.

3) $m - g = \frac{x+y}{2} - \sqrt{x.y} = \frac{1}{2} ((\sqrt{x})^2 + (\sqrt{y})^2 - 2\sqrt{x.y}) = \frac{1}{2} (\sqrt{y} - \sqrt{x})^2 \geq 0 \implies m - g \geq 0 \implies g \leq m$.

So, $x \leq g \leq m \leq y$.

4) According 1), the arithmetic mean to $\frac{1}{x}$ is between $\frac{1}{x}$ and $\frac{1}{y}$, that's provide

$$\frac{1}{y} \leq \frac{1}{h} \leq \frac{1}{x} \implies x \leq h \leq y.$$

5) According 3), the geometric mean to two real numbers $\frac{1}{x}$ and $\frac{1}{y}$ is less than or equals to their

arithmetic mean, that's provide $\sqrt{\frac{1}{x} \cdot \frac{1}{y}} \leq \frac{1}{2} \left(\frac{1}{x} + \frac{1}{y} \right) \implies \frac{1}{g} \leq \frac{1}{h} \implies h \leq g$.

Finally, we proved that $x \leq h \leq g \leq m \leq y$.

Exercise 2 : Calculate the sum :

- 1- S_n of the first n odd integers.
- 2- $S = 1.2 + 2.3 + 3.4 + \dots + (n-1).n$
- 3- $U = 1.1! + 2.2! + 3.3! + \dots + n.n!$
- 4- $T = 1.n + 2.(n-1) + \dots + (n-1).2 + n.1$
- 5- $V = 1.2.3 + 2.3.4 + 3.4 + \dots + (n-2).(n-1).n$

Knowing the expressions of $\sum_{k=1}^n k$ and $\sum_{k=1}^n k^2$

Solution :

- 1- The successive odd integers are $u_k = 2k-1$ with $k = 1, 2, \dots$. So, the n -th odd integer is $u_n = 2n-1$.

We are asked to calculate $S_n = u_1 + u_2 + \dots + u_n$

We check that $S_1 = 1$, $S_2 = 1 + 3 = 4$, $S_3 = 1 + 3 + 5 = 9, \dots$, therefore it seems that the formula sought is $S_n = n^2$, if it true at the rank n . So,

$$\begin{aligned} S_{n+1} = u_1 + u_2 + \dots + u_n + u_{n+1} &= S_n + u_{n+1} \\ &= S_n + (2n+1) \quad (\text{because } u_{n+1} = 2(n+1) - 1 = 2n+2-1 = 2n+1) \\ &= n^2 + 2n + 1 = (n+1)^2 \end{aligned}$$

Conclusion : For all integer n , $S_n = n^2$.

- 2- We can find S Knowing the sums $\sum_{k=1}^n k$ and $\sum_{k=1}^n k^2$.

$$\begin{aligned} S = \sum_{k=1}^n (k-1)k &= \sum_{k=1}^n (k^2 - k) = \sum_{k=1}^n k^2 - \sum_{k=1}^n k = \frac{1}{6}n(n+1)(2n+1) - \frac{1}{2}n(n+1) \\ &= \frac{1}{6}n(n+1)((2n+1) - 3) \\ &= \frac{1}{6}n(n+1)((2n-2)) \\ &= \frac{1}{6}n(n+1)(2(n-1)) \\ &= \frac{1}{3}n(n+1)(n-1) \end{aligned}$$

second method : We pose $v_k = k(k-1)(k-2)$, we remark that

$$v_{k+1} - v_k = (k+1)k(k-1) - k(k-1)(k-2) = 3k(k-1). \text{ So,}$$

$$\begin{aligned}
 S = \sum_{k=1}^n (k-1)k &= \frac{1}{3} \sum_{k=1}^n (v_{k+1} - v_k) = \frac{1}{3} ((v_2 - v_1) + (v_3 - v_2) + \dots + (v_{n+1} - v_n)) \\
 &= \frac{1}{3} (v_{n+1} - v_1) \\
 &= \frac{1}{3} n(n+1)(n-1).
 \end{aligned}$$

3- We observe that $k.k! = (k+1-1).k! = (k+1).k! - k! = (k+1)! - k!$. So, we deduce that

$$\begin{aligned}
 U = \sum_{k=1}^n [(k+1)! - k!] &= [(2! - 1!) + (3! - 2!) + \dots + ((n+1)! - n!)] \\
 &= (n+1)! - 1! = (n+1)! - 1.
 \end{aligned}$$

4- We can be used two methods at least as in previous calculation

$$\begin{aligned}
 T = \sum_{k=1}^n k(n+1-k) &= \sum_{k=1}^n [k(n+1) - k^2] \\
 &= (n+1) \sum_{k=1}^n k - \sum_{k=1}^n k^2 \\
 &= (n+1) \frac{1}{2} n(n+1) - \frac{1}{6} n(n+1)(2n+1) \\
 &= \frac{1}{6} n(n+1) [3(n+1) - (2n+1)] \\
 &= \frac{1}{6} n(n+1)(n+2).
 \end{aligned}$$

Second method : We pose $u_k = k(n+1-k)$, $v_k = k(k-1)(k-2)$ and $w_k = k(k-1)$. We have

$$\begin{aligned}
 v_{k+1} - v_k &= (k+1)k(k-1) - k(k-1)(k-2) = k(k-1)[k+1-k+2] \\
 &= 3k(k-1)
 \end{aligned}$$

and

$$w_{k+1} - w_k = (k+1)k - k(k-1) = k(k+1-k+1) = 2k$$

We deduce that $u_k = a_{k+1} - a_k$, with $a_k = \frac{n}{2}w_k - \frac{1}{3}v_k$. So,

$$\begin{aligned}
 T = \sum_{k=1}^n a_{k+1} - a_k &= (a_2 - a_1) + (a_3 - a_2) + \dots + (a_{n+1} - a_n) \\
 &= a_{n+1} - a_1 = a_{n+1} \\
 &= \frac{n}{2}w_{n+1} - \frac{1}{3}v_{n+1} \\
 &= \frac{n}{2}[n(n+1)] - \frac{1}{3}[n(n+1)(n-1)] \\
 &= \frac{n^2}{2}(n+1) - \frac{1}{3}n(n+1)(n-1) \\
 &= \frac{1}{6}n(n+1)[3n - 2(n-1)] \\
 &= \frac{1}{6}n(n+1)(n+2).
 \end{aligned}$$

5- We pose $u_k = k(k+1)(k+2)$ and $v_k = (k-1)k(k+1)(k+2)$, we observe that,

$$\begin{aligned}
 v_{k+1} - v_k &= k(k+1)(k+2)(k+3) - (k-1)k(k+1)(k+2) = k(k+1)(k+2)[k+3 - k+1] \\
 &= 4k(k+1)(k+2) = 4u_k.
 \end{aligned}$$

We deduce that

$$\begin{aligned}
 V = \sum_{k=1}^{n-2} u_k &= \frac{1}{4} \sum_{k=1}^{n-2} (v_{k+1} - v_k) = \frac{1}{4} [(v_2 - v_1) + (v_3 - v_2) + \dots + (v_{n-1} - v_{n-2})] \\
 &= \frac{1}{4}(v_{n-1} - v_1) \\
 &= \frac{1}{4}v_{n-1} \\
 &= \frac{1}{4}(n-2)(n-1)n(n+1)
 \end{aligned}$$

Exercise 3 : Given A and B , two non-empty and bounded sets of $\mathbb{R}$, show that :

1- $\sup(A \cup B) = \max\{\sup A, \sup B\}$

2- $\inf(A \cup B) = \min\{\inf A, \inf B\}$.

Consider the set of $\mathbb{R}$, $X = \left\{ (-1)^n + \frac{1}{n} \right\}$

Show that X has a supremum and infimum, determine maximum and minimum if they exist.

Solution :

1- **Prove that** $\inf(A \cup B) = \min\{\inf A, \inf B\}$

We have $A \subset (A \cup B) \implies \inf A \geq \inf(A \cup B)$ and $B \subset (A \cup B) \implies \inf B \geq \inf(A \cup B)$. So,

$$\inf(A \cup B) \leq \min\{\inf A, \inf B\} \tag{1.1}$$

Let $x \in (A \cup B) \implies x \in A$ or $x \in B$

If $x \in A \implies x \geq \inf A \geq \min\{\inf A, \inf B\}$, If $x \in B \implies x \geq \inf B \geq \min\{\inf A, \inf B\}$. So, $x \geq \min\{\inf A, \inf B\}$

In other words $\min\{\inf A, \inf B\}$ is an lower bound of $(A \cup B)$ and $\inf(A \cup B)$ is the largest of lower bound, so

$$\inf(A \cup B) \geq \min\{\inf A, \inf B\} \quad (1.2)$$

Conclusion : in (1.1) and (1.2) we show that $\inf(A \cup B) = \min\{\inf A, \inf B\}$

2- **Prove that** $\sup(A \cup B) = \max\{\sup A, \sup B\}$

We have $A \subset (A \cup B) \implies \sup A \leq \sup(A \cup B)$ and $B \subset (A \cup B) \implies \sup B \leq \sup(A \cup B)$. So,

$$\sup(A \cup B) \geq \max\{\sup A, \sup B\} \quad (1.3)$$

Let $x \in (A \cup B) \implies x \in A$ or $x \in B$

If $x \in A \implies x \leq \sup A \leq \max\{\sup A, \sup B\}$, If $x \in B \implies x \leq \sup B \leq \max\{\sup A, \sup B\}$. So, $x \leq \max\{\sup A, \sup B\}$

In other words $\max\{\sup A, \sup B\}$ is an upper bound of $(A \cup B)$ and $\sup(A \cup B)$ is the smallest of upper bound, so

$$\sup(A \cup B) \leq \max\{\sup A, \sup B\} \quad (1.4)$$

Conclusion : in (1.3) and (1.4) we show that $\sup(A \cup B) = \max\{\sup A, \sup B\}$

2- We consider the set of $\mathbb{R}$, $X = \{(-1)^n + \frac{1}{n}\}$.

The sequence $u_n = (-1)^n + \frac{1}{n}$ niether increasing ni decreasing, is the alternating sign, we consider two sequences $(v_n)_{n \geq 1}$ and $(w_n)_{n \geq 0}$ of a real numbers defined by

$$v_n = u_{2n} = 1 + \frac{1}{2n} \text{ and } w_n = u_{2n+1} = -1 + \frac{1}{2n+1}$$

For all $n \geq 1$, $v_{n+1} - v_n = 1 + \frac{1}{2n+2} - 1 - \frac{1}{2n} = \frac{-2}{2n(2n+2)}$.

So, the sequence $(v_n)_{n \geq 1}$ is decreasing and we have

$$v_1 = \frac{3}{2}, \lim_{n \rightarrow \infty} v_n = 1.$$

For all $n \geq 0$, $w_{n+1} - w_n = -1 + \frac{1}{2n+3} + 1 - \frac{1}{2n+1} = \frac{-2}{(2n+1)(2n+2)}$.

So, the sequence $(w_n)_{n \geq 0}$ is decreasing and we have

$$w_0 = 0, \lim_{n \rightarrow \infty} w_n = -1.$$

Finally,

$$X = \left\{ 1 + \frac{1}{2n}, n \in \mathbb{N}^* \right\} \cup \left\{ -1 + \frac{1}{2n+1}, n \in \mathbb{N} \right\}$$

So, $\sup X = \max\{\sup A, \sup B\} = \max\{\frac{3}{2}, 0\} = \frac{3}{2}$ and $\inf X = \min\{\inf A, \inf B\} = \min\{1, -1\} = -1$.

Exercise 4 : Let n be a natural number. Calculate the sums :

A- $C_n^0 + 2C_n^2 + \dots + 2^p C_n^{2p} + \dots$

B- $C_n^1 + 2C_n^3 + \dots + 2^p C_n^{2p+1} + \dots$

C- $C_n^1 + 2C_n^2 + \dots + nC_n^n = \sum_{k=1}^n kC_n^k$

D- $C_n^0 + \frac{1}{2}C_n^1 + \frac{1}{3}C_n^2 + \dots + \frac{1}{n+1}C_n^n = \sum_{k=0}^n \frac{1}{k+1}C_n^k$.

solution :

1- and 2- We use the development of $(1+x)^n$ (Newton binomial) with $x = \pm\sqrt{2}$, we find

$$\begin{aligned} (1 + \sqrt{2})^n &= \sum_{k=0}^n (\sqrt{2})^k C_n^k \\ &= C_n^0 + \sqrt{2}C_n^1 + 2C_n^2 + 2\sqrt{2}C_n^3 + 2^2C_n^4 + 2^2\sqrt{2}C_n^5 + \dots \\ &= C_n^0 + 2C_n^2 + 2^2C_n^4 + \dots + (C_n^1 + 2C_n^3 + 2^2C_n^5 + \dots) \sqrt{2} \\ &= A + B\sqrt{2}. \end{aligned}$$

The same, $(1 - \sqrt{2})^n = A - B\sqrt{2}$, so

$$\begin{cases} (1 + \sqrt{2})^n = A + B\sqrt{2} & \dots 1, \\ (1 - \sqrt{2})^n = A - B\sqrt{2} & \dots 2. \end{cases}$$

$$1+2 \iff 2A = [(1 + \sqrt{2})^n + (1 - \sqrt{2})^n] \iff A = \frac{1}{2} [(1 + \sqrt{2})^n + (1 - \sqrt{2})^n]$$

$$1-2 \iff 2\sqrt{2}B = [(1 + \sqrt{2})^n - (1 - \sqrt{2})^n] \iff B = \frac{1}{2\sqrt{2}} [(1 + \sqrt{2})^n - (1 - \sqrt{2})^n]$$

3- **First method :** We have $C = \sum_{k=1}^n kC_n^k$, such that

$$kC_n^k = k \frac{n!}{k!(n-k)!} = \frac{n!}{(k-1)!(n-k)!} = n \frac{(n-1)!}{k!(n-k)!} = nC_{n-1}^{k-1}.$$

We deduce that

$$\begin{aligned} C &= n \sum_{k=1}^n C_{n-1}^{k-1} = n \sum_{k=0}^{n-1} C_{n-1}^k \\ &= n(C_{n-1}^0 + C_{n-1}^1 + \dots + C_{n-1}^{n-1}) \\ &= n2^{n-1} \end{aligned}$$

Second method : For all real number x , we know that $(1+x)^n = \sum_{k=0}^n C_n^k x^k$, if we derive this equality with respect to x we find

$$\forall x \in \mathbb{R}, n(1+x)^{n-1} = \sum_{k=1}^n k C_n^k x^{k-1}$$

especially with $x = 1$ we obtain $\sum_{k=1}^n k C_n^k = n2^{n-1}$

4- As for C we denote that

$$\begin{aligned} (n+1)C_n^k &= \frac{(n+1).n!}{k!(n-k)!} = \frac{(n+1).n! \times (k+1)}{k!(n-k)! \times (k+1)} \\ &= (k+1) \frac{(n+1)!}{(k+1)!(n-k)!} \\ &= (k+1)C_{n+1}^{k+1}. \end{aligned}$$

So, $\frac{1}{k+1}C_n^k = \frac{1}{n+1}C_{n+1}^{k+1}$, thus,

$$\begin{aligned} D &= \sum_{k=0}^n \frac{1}{k+1} C_n^k = \sum_{k=0}^n \frac{1}{n+1} C_{n+1}^{k+1} = \frac{1}{n+1} \sum_{k=0}^n C_{n+1}^{k+1} \\ &= \frac{1}{n+1} (2^{n+1} - 1). \end{aligned}$$

Second method : We integrate the equality $(1+x)^n = \sum_{k=0}^n C_n^k x^k$ between 0 and x , we find

$$\begin{aligned} \int_0^x (1+s)^n ds &= \int_0^x \sum_{k=0}^n C_n^k s^k ds \iff \frac{1}{n+1} [(1+s)^{n+1}]_0^x = \sum_{k=0}^n C_n^k \int_0^x s^k ds \\ &\iff \frac{1}{n+1} [(1+x)^{n+1} - 1] = \sum_{k=0}^n C_n^k \left[\frac{1}{k+1} s^{k+1} \right]_0^x \\ &\iff \frac{1}{n+1} [(1+x)^{n+1} - 1] = \sum_{k=0}^n C_n^k \frac{1}{k+1} x^{k+1}. \end{aligned}$$

With $x = 1$ we obtain

$$\sum_{k=0}^n \frac{1}{k+1} C_n^k = \frac{1}{n+1} (2^{n+1} - 1).$$

Exercise 5 :

Calculate $\lim_{n \rightarrow +\infty} \frac{1}{n^2} ([x] + [2x] + \dots + [nx])$

Let x be a real number, and let n be a non-zero natural number. We denote $t \rightarrow [t]$ “the integer part”.

Show that for every integer k , $\left[\frac{x+k}{n} \right] = \left[\frac{[x]+k}{n} \right]$.

Solution : For all $x \in \mathbb{R}$ and all integer k we have $kx - 1 \leq [kx] \leq kx$ (because $x - 1 \leq [x] \leq x$).

$$\frac{1}{n^2} ([x] + [2x] + \dots + [nx]) = \frac{1}{n^2} \sum_{k=1}^n [kx] = u_n(x)$$

So,

$$\begin{aligned} kx - 1 \leq [kx] \leq kx &\iff \sum_{k=1}^n (kx - 1) \leq \sum_{k=1}^n [kx] \leq \sum_{k=1}^n kx \\ &\iff \frac{1}{n^2} \sum_{k=1}^n (kx - 1) \leq \frac{1}{n^2} \sum_{k=1}^n [kx] \leq \frac{1}{n^2} \sum_{k=1}^n kx \\ &\iff \frac{x}{n^2} \sum_{k=1}^n k - \frac{1}{n^2} \sum_{k=1}^n 1 \leq u_n(x) \leq \frac{x}{n^2} \sum_{k=1}^n k \\ &\iff \frac{x}{n^2} \frac{n(n+1)}{2} - \frac{1}{n} \leq u_n(x) \leq \frac{x}{n^2} \frac{n(n+1)}{2} \quad (\text{because } \sum_{k=1}^n 1 = n) \\ &\iff x \frac{n^2 + n}{2n^2} - \frac{1}{n} \leq u_n(x) \leq x \frac{n^2 + n}{2n^2} \end{aligned}$$

We pass to the limite, we obtain $\lim_{n \rightarrow +\infty} u_n(x) = \frac{x}{2}$.

2. Either $q = \left\lfloor \frac{x+k}{n} \right\rfloor$. We have

$$q \leq \frac{x+k}{n} < q+1 \iff qn \leq x+k < qn+n \iff qn-k \leq x < qn+n-k$$

For all integers m and p we have, $m \leq x < p \iff m \leq [x] < p$. So,

$$qn-k \leq [x] < qn+n-k \iff qn \leq [x] + k < qn+n \iff q \leq \frac{[x] + k}{n} < q+1$$

$$\text{So, } \left\lfloor \frac{x+k}{n} \right\rfloor = \left\lfloor \frac{[x] + k}{n} \right\rfloor.$$

Exercise 6 : Let $x_1, x_2, \dots, x_n$ be a family of real numbers in $[0, 1]$.

Prove the inequality $\prod_{k=1}^n (1 - x_k) \geq 1 - \sum_{k=1}^n x_k$.

remark :

$$\begin{aligned} \prod_{k=1}^n (1 - x_k) &= (1 - x_1)(1 - x_2) \dots (1 - x_n) \\ \sum_{k=1}^n x_k &= x_1 + x_2 + \dots + x_n \end{aligned}$$

Solution : Let $x_1, x_2, \dots, x_n$ be a family of real numbers in $[0, 1]$, We prove by induction about the integer $n \geq 1$

- $n = 1$: $P(1) : (1 - x_1) \geq (1 - x_1)$ (True).
- $n = 2$: $P(2) :$

$$\begin{aligned}
 \prod_{k=1}^2 (1 - x_k) &= (1 - x_1)(1 - x_2) = 1 - x_2 - x_1 + x_1x_2 = 1 - (x_1 + x_2) + x_1x_2 \\
 &\geq 1 - (x_1 + x_2) \\
 &\geq 1 - \sum_{k=1}^2 x_k \quad (True).
 \end{aligned}$$

- $P(n + 1)$: We assume that $P(n)$ is true and we show that $P(n + 1)$ is true.

$$\begin{aligned}
 \prod_{k=1}^{n+1} (1 - x_k) &= (1 - x_1)(1 - x_2) \dots (1 - x_n)(1 - x_{n+1}) \\
 &= (1 - x_{n+1}) \prod_{k=1}^n (1 - x_k) = (1 - x_{n+1})(1 - x_2) \dots (1 - x_n) \\
 &\geq (1 - x_{n+1}) \left(1 - \sum_{k=1}^n x_k \right) \\
 &\geq 1 - \sum_{k=1}^n x_k - x_{n+1} + x_{n+1} \sum_{k=1}^n x_k \\
 &\geq 1 - \left(x_{n+1} + \sum_{k=1}^n x_k \right) + x_{n+1} \sum_{k=1}^n x_k \\
 &\geq 1 - \sum_{k=1}^{n+1} x_k + x_{n+1} \sum_{k=1}^n x_k \\
 &\geq 1 - \sum_{k=1}^{n+1} x_k \quad (True).
 \end{aligned}$$

Conclusion : For all $n \geq 1$: $\prod_{k=1}^n (1 - x_k) \geq 1 - \sum_{k=1}^n x_k$.

Some calculations details concerning exercise 4

Show that $\sum_{k=0}^n C_n^k = 2^n$, $\sum_{k=0}^{n-1} C_{n-1}^k = 2^{n-1}$, and $\sum_{k=0}^n C_{n+1}^{k+1} = 2^{n+1} - 1$.

Newton binomial : $(x + y)^n = \sum_{k=0}^n C_n^k x^k y^{n-k}$.

1. We use the newton binomial with $x = y = 1$

$$\sum_{k=0}^n C_n^k = (1 + 1)^n = 2^n.$$

2. $\sum_{k=0}^{n-1} C_{n-1}^k$. We pose $N = n - 1$ we obtain

$$\sum_{k=0}^{n-1} C_{n-1}^k = \sum_{k=0}^{n-1} C_N^k = 2^{n-1}.$$

3. $\sum_{k=0}^n C_{n+1}^{k+1}$. We pose $N = n + 1$ and $K = k + 1$,

$$k = 0 \implies K = 0 + 1 = 1, \text{ so } \sum_{K=1}^n C_N^K = \sum_{K=0}^n C_N^K - C_N^0$$

$$(\text{because } \sum_{K=0}^n C_N^K = C_N^0 + C_N^1 + C_N^2 + \dots + C_N^N \implies \sum_{K=0}^n C_N^K = C_N^0 + \sum_{K=1}^n C_N^K)$$

$$\text{implies that } \sum_{K=1}^n C_N^K = \sum_{K=0}^n C_N^K - C_N^0 = 2^N - 1 = 2^{n+1} - 1$$

$$\text{So, } \sum_{k=0}^n C_{n+1}^{k+1} = 2^{n+1} - 1.$$